SAGAR CHAUDHARY

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SUMMARY

AI Engineer specializing in Generative AI and agentic systems. Builds production RAG pipelines end to end - chunking strategy, embedding selection (Nomic, Gemini), ChromaDB vector stores, cross-encoder reranking and prompt-to-template design - and stateful multi-step agents with LangGraph and MCP that route between tools for natural-language-to-SQL and automated report generation. Serves LLMs across Ollama, OpenAI and Gemini behind a streaming FastAPI layer. Backs it with real deep-learning depth (PyTorch, LSTM, CNNs, BERT) and production MLOps: Docker, Kubernetes with KEDA autoscaling, CI/CD, Prometheus and OpenTelemetry observability.

TECHNICAL SKILLS

Generative AI & Agentic: LangChain, LangGraph, RAG, agentic AI (tool-use agents, MCP), chunking strategies, prompt engineering, prompt-to-template design, rerankers, vector search (ChromaDB), LLM & embedding APIs (OpenAI, Gemini, Ollama, Nomic)

Deep Learning: PyTorch, TensorFlow/Keras, CNNs (YOLO, Detectron2, ZoeDepth), LSTM/RNN time-series models, BERT/Transformers, transfer learning, LiDAR-camera sensor fusion, OpenCV, Open3D

Machine Learning: classification, regression, time-series forecasting (LSTM, ARIMA), ensembles (XGBoost, LightGBM, AdaBoost), feature engineering, data cleaning & pre-processing, SMOTE, Optuna, Scikit-Learn, Pandas, NumPy, Statsmodels

MLOps & Deployment: Docker, Kubernetes, KEDA autoscaling, CI/CD (GitHub Actions), Prometheus, OpenTelemetry, ONNX Runtime model serving, structured logging, monitoring, crash-recovery design

Backend & Data Pipelines: Python, Go, Java, FastAPI, REST APIs, WebSockets, TCP/IP sockets, microservices, event-driven pub-sub, Redis Streams, asyncio, JWT, RBAC

Databases & Cloud: PostgreSQL (SQL), NoSQL (MongoDB), MySQL, ChromaDB, AWS, Linux, Git

Core CS: Data Structures, OOP, Distributed Systems, System Design, Concurrency, Operating Systems, Computer Networks

EXPERIENCE

Deevia Software, Bengaluru — **Apr 2025 - Present**

Graduate Trainee Engineer — *Nov 2025 - Present*

Real-Time Engine Test-Bed Monitoring & Telemetry Platform (Industrial IT-OT System)

Tech Stack: Python, FastAPI, LangChain, LangGraph, LSTM (PyTorch), RAG, ChromaDB, Ollama, OpenAI, Gemini, WebSockets, TCP/IP Sockets, PostgreSQL, React.js, TypeScript, Redux Toolkit, Docker, JWT, RBAC

- Architected an end-to-end IT-OT data pipeline for an engine test facility - ingesting high-frequency industrial sensor data over raw TCP/IP sockets and streaming it to live dashboards via WebSockets at sub-second latency.
- Trained LSTM deep-learning models on engine sensor time-series for trend forecasting, anomaly detection, and predictive-maintenance insights during live test runs.
- Built a LangChain-based RAG assistant over test reports - designed the chunking strategy, prompt-to-template pipeline, Nomic/Gemini embeddings into a ChromaDB vector store, and a reranking stage serving LLMs via Ollama, OpenAI, and Gemini - enabling natural-language querying of historical test data and eliminating manual report lookups.
- Developed agentic AI workflows with LangGraph - a stateful multi-step agent that routes between tools for natural-language-to-SQL querying of test data and automated test-report generation and summarization.
- Optimized ingestion with asynchronous processing and multithreaded batched database writes, sustaining continuous high-volume sensor load without memory backpressure.
- Designed an event-driven publish-subscribe architecture that decoupled ingestion, processing, and streaming services, so new sensor types are added through configuration rather than code changes.

Multi-Vendor Camera Device Management Framework

Tech Stack: Python, FastAPI, PostgreSQL, MongoDB (NoSQL), React, Docker, CI/CD, OOP Design Patterns

- Built a device management framework using object-oriented design - abstract base classes and interface layers - to abstract camera hardware, enabling drop-in integration of multiple camera vendors without backend changes.
- Designed a dual-database layer - PostgreSQL for relational device records and MongoDB (NoSQL) for heterogeneous, schema-less device configurations - to handle structured and unstructured data efficiently.

Machine Learning Intern — *Apr 2025 - Nov 2025*

Distributed Multi-Sensor Video Analytics Platform

Tech Stack: Go, Python, FastAPI, REST APIs, WebSockets, AWS, PostgreSQL, Docker, Microservices, OpenCV

- Designed a distributed microservices architecture on AWS separating ingestion, processing, and streaming layers for scalable multi-camera plus sensor pipelines.
- Built a dual-backend system: Go services handling real-time API orchestration and request validation, with Python services running compute-intensive video processing.

3D Object Detection via LiDAR-Camera Sensor Fusion (KITTI)

Tech Stack: Python, PyTorch, YOLOv8 (CNN), SFA3D, OpenCV, Open3D, NumPy, SLAM, RANSAC/PROSAC, Docker

- Built a multi-modal 3D object detection system fusing LiDAR point-cloud detections (SFA3D) with CNN-based 2D camera detections (YOLOv8) for robust perception in autonomous-driving scenes.
- Designed a confidence-weighted fusion algorithm that combines per-sensor detections by IoU association, improving accuracy and robustness over single-sensor baselines - 92.44% / 80.23% AP (easy / hard) on the KITTI benchmark.

Foot Monitoring System (ADAS)

Tech Stack: Python, PyTorch, YOLOv11, Detectron2, ZoeDepth, OpenCV, FastAPI, React, PostgreSQL, AWS

- Built a production computer-vision system detecting driver pedal interactions (accelerator, brake, clutch) by combining segmentation, keypoint detection, and object detection deep-learning models.
- Developed pedal-position estimation and monocular depth inference using ZoeDepth with custom logic to classify pedal depression and distinct foot actions.

PROJECTS

Inferno - Distributed ML Inference & Agentic AI Platform (Live Demo · GitHub)

Tech Stack: Python, FastAPI, Redis (Streams + Pub/Sub), WebSockets, asyncio, PyTorch, ONNX Runtime, YOLOv8, faster-whisper, Sentence-Transformers (RAG), MCP (Agentic AI), Prometheus, OpenTelemetry, React, TypeScript, Docker, Kubernetes, KEDA, GitHub Actions

- Built a horizontally-scalable model-serving platform - a stateless async FastAPI gateway, a Redis-Streams broker, and a pool of shared-nothing worker processes - that batches inference requests and streams results back over WebSockets to a live ops dashboard, solving the core serving problem of high throughput and low-latency real-time delivery.
- Shipped an agentic AI layer: an MCP agent server exposing platform tools to LLM agents (tool-use and API collaboration), streaming local-LLM chat (SSE), and a RAG pipeline with bi-encoder retrieval, cross-encoder reranking, and cited answers.
- Implemented dynamic request batching (size-or-timeout window) that amortizes per-call model overhead and keeps the GPU/CPU saturated - raising throughput under load while capping tail latency at the batch-wait window, with no manual tuning.
- Engineered backpressure with hysteresis (HTTP 429 + Retry-After, dual high/low water-marks) so the platform sheds load and stays stable under overload instead of collapsing into latency spikes and OOM.
- Generalized the system with a config-driven, model-agnostic plugin registry serving 7 model types - YOLOv8 detection, Whisper transcription, ONNX/HuggingFace classification, and RAG question-answering.

Disaster Tweet Classification

Tech Stack: Python, NLTK, Scikit-Learn, TensorFlow/Keras, BERT, TF-IDF

- Collected real-time tweet data via the Twitter API and preprocessed noisy social text with tokenization, lemmatization, and stopword removal to build a clean training corpus.
- Benchmarked standalone TF-IDF and BERT models, then built a hybrid classifier combining BERT embeddings with TF-IDF features - 92.8% accuracy / 91% F1, outperforming single-model baselines.

Customer Churn Prediction

Tech Stack: Python, Pandas, Scikit-Learn, XGBoost, LightGBM, AdaBoost, Optuna, SMOTE

- Handled class imbalance with SMOTE and tuned hyperparameters with Optuna - 94.6% accuracy, 0.96 AUC-ROC, 93.2% precision.

ACHIEVEMENTS & CERTIFICATIONS

- Top 10 Finalist - HumanAIze YouData.ai Hackathon (out of 150 teams); built a scalable NLP pipeline for document classification. Hack2skill certificate (ID: 2024H2S06EH10008).
- 5th Place - CODE RED 2024 Hackathon (24-hour event); led a team of 4 to build a marine-life classification system using CNNs, real-time image processing, and transfer learning.
- Machine Learning Specialization - Stanford / DeepLearning.AI (Coursera), Dec 2023
- Neural Networks and Deep Learning - DeepLearning.AI, May 2024 · OpenCV Bootcamp - OpenCV University, Apr 2025 (Grade: 100%)

EDUCATION

Bangalore Institute of Technology — B.E. in Information Science and Engineering (CGPA: 8.6) **Dec 2021 - Jul 2025**

DAV Public School, Jamshedpur — CBSE Class 12: 92.4% **Apr 2020 - May 2021**

DAV Public School, Jamshedpur — CBSE Class 10: 92.4% **Apr 2018 - Apr 2019**